

LAB 6

TECH 60 - Sec 14

F26

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Oct 7, 2026

Objective:

The objective of this lab is to understand how voltage and current behave in a parallel circuit and to confirm the Current Divider Rule and Kirchhoff's Current Law (KCL). We will do this by comparing hand calculations, a Multisim simulation and breadboard measurements.

Equipment/Materials:

Multisim software running on PC

330 Ω resistors (Quantity 1)

470 Ω resistors (Quantity 1)

1 k Ω resistors (Quantity 1)

Part I. Hand Calculations

$$V_S = 9 \text{ V}$$

$$R_1 = 330 \Omega$$

$$R_2 = 470 \Omega$$

$$R_3 = 1 \text{ k}\Omega = 1000 \Omega$$

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

$$1/R_T = 1/330 \Omega + 1/470 \Omega + 1/1000 \Omega$$

$$1/R_T = 0.0030303 \text{ S} + 0.0021277 \text{ S} + 0.0010000 \text{ S} = 0.0061580 \text{ S}$$

$$R_T = 1/0.0061580 \text{ S}$$

$$R_T = 162.39 \Omega$$

$$I_T = V_S/R_T$$

$$I_T = 9 \text{ V}/162.39 \Omega$$

$$I_T = 0.055422 \text{ A} = 55.422 \text{ mA}$$

$$I_n = I_T * (R_T/R_n)$$

$$I_1 = I_T * (R_T/R_1)$$

$$I_1 = 55.422 \text{ mA} * (162.39 \Omega/330 \Omega)$$

$$I_1 = 27.273 \text{ mA}$$

$$I_2 = I_T * (R_T / R_2)$$

$$I_2 = 55.422 \text{ mA} * (162.39 \Omega / 470 \Omega)$$

$$I_2 = 19.149 \text{ mA}$$

$$I_3 = I_T * (R_T / R_3)$$

$$I_3 = 55.422 \text{ mA} * (162.39 \Omega / 1000 \Omega)$$

$$I_3 = 9.000 \text{ mA}$$

$$I_T = I_1 + I_2 + I_3$$

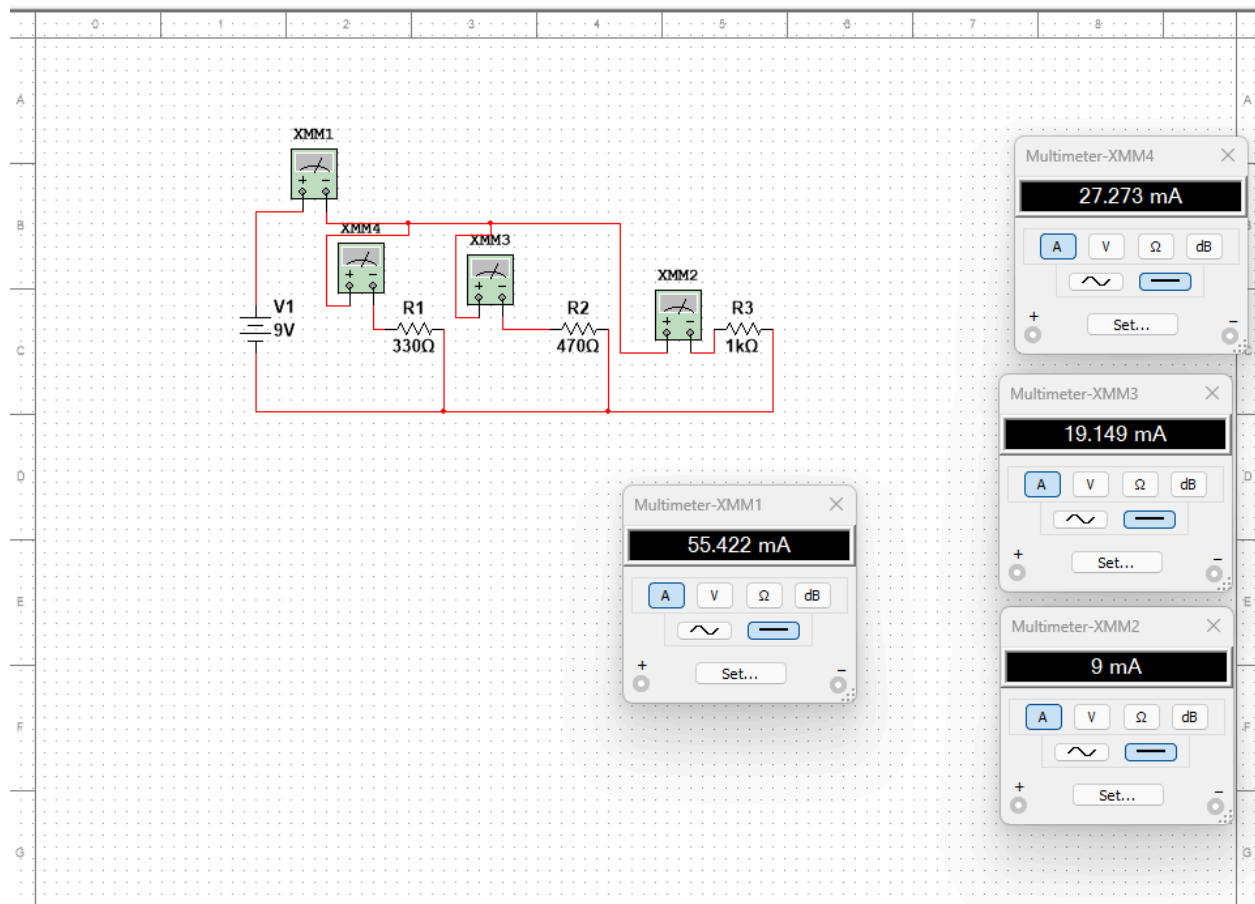
$$I_1 + I_2 + I_3 = 27.273 \text{ mA} + 19.149 \text{ mA} + 9.000 \text{ mA}$$

$$I_1 + I_2 + I_3 = 55.422 \text{ mA}$$

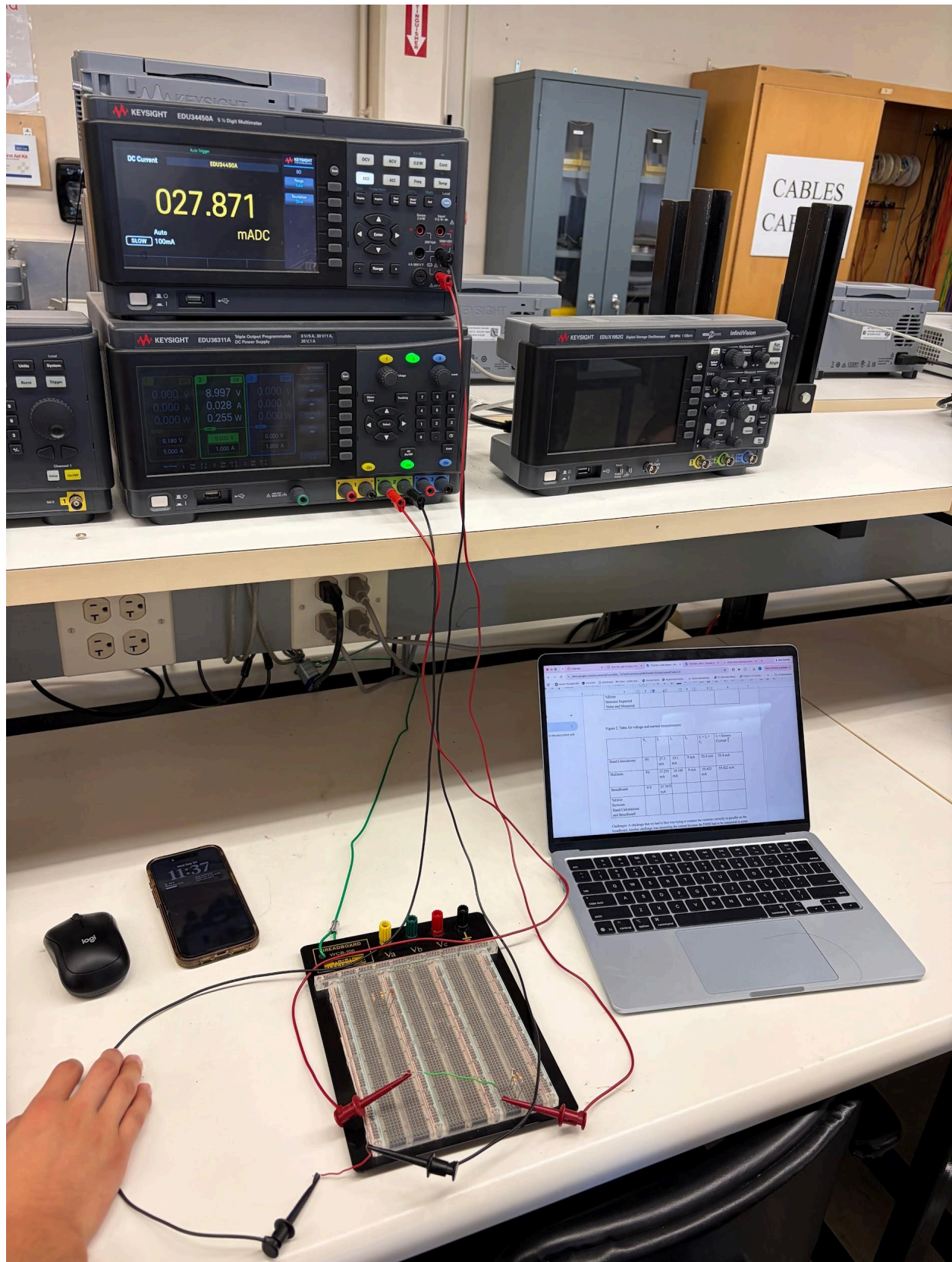
$$I_T = 55.422 \text{ mA}$$

$$55.422 \text{ mA} = 55.422 \text{ mA} \quad \text{KCL confirmed}$$

Part II. Circuit simulation using Multisim



Part III. Circuit construction on the breadboard







Part IV. Questions

- 1. Compare the measured resistor values to the expected values. If they are not exactly the same, then explain briefly why they are not exactly equal.**

The measured values are not exactly the same as what we expected but they are very close. We were within 0.30% error. Also real resistors are made with tolerance ratings so this is expected. The DMM also has its own accuracy limit and the probe contact as well as human pressure and error could be a part of the error.

- 2. What do you observe regarding the relationship between I_1 , I_2 , I_3 in comparison to the total current, I_T ?**

The three branch currents add up to the total current. In the hand calculations this added to 55.422 mA. In the real circuit, we found it was at 56.188 mA which is still very close given our percent error.

- 3. Write a paragraph about the relationship between the current in each resistor in relation to their corresponding resistor value.**

Each resistor is connected directly across the source, so all three resistors have the same 9 volts across them. Using Ohm's law tells us that $V/R = I$. We found that the smallest resistor carried a current of 27.273 mA, R_2 carried 19.149 mA, and R_3 carried 9mA. The ratios of the values confirm the theory that $I_T = I_1 + I_2 + I_3$. In a parallel circuit, the current takes the easiest path and splits unevenly, with the lowest resistance getting the most current.

- 4. Compare the measured total current to the calculated, and the Multisim value of this quantity. Should they be the same value ideally? Explain briefly why they are not exactly equal.**

The values for current between the hand calculations and multisim are very close since they are both theoretical and are either given or inputted. The physical resistors did stray a little from the hand calculated/multisim value since over time equipment can deteriorate and have a percent error ranging from 5% of the resistance value. The highest percent error we encountered was 1.79% with the current across R_1 .

- 5. From your experimental results have you confirmed the Kirchhoff's Current Law (KCL)? Explain**

Yes, KCL was confirmed through all methods that were conducted during the experiment, including hand calculations, multisim, and breadboard values. With the breadboard and hand calculated values having a percent error of 0.72%, confirming $I_T = I_1 + I_2 + I_3$.

Figure 1. Resistance of each resistor

	R_1	R_2	R_3	R_T
Expected value	330 ohms	470 ohms	1k ohms	162.4 ohms
Measured	331 ohms	468 ohms	983 ohms	161.94 ohms

%Error Between Expected Value and Measured	0.3%	0.43%	1.7%	0.28%
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Figure 2. Table for voltage and current measurements:

	V_s	I_1	I_2	I_3	$I_1 + I_2 + I_3$	$I_T = \text{Source Current}$
Hand Calculations	9V	27.3 mA	19.1 mA	9 mA	55.4 mA	55.4 mA
Multisim	9V	27.273 mA	19.149 mA	9 mA	55.422 mA	55.422 mA
Breadboard	9 V	27.79 mA	19.301 mA	9.097 mA	56.188 mA	55 mA
%Error Between Hand-Calculations and Breadboard	0.00%	1.79%	1.05%	1.08%	1.42%	0.72%

Challenges: A challenge that we had to face was trying to connect the resistors correctly in parallel on the breadboard. Another challenge was measuring the current because the DMM had to be connected in series with the circuit and set to the correct current setting. Our calculations were slightly different when we were measuring the resistors and current values. Overall, checking the circuit connections and making sure the DMM was connected correctly helped us obtain more accurate measurements.

Conclusion: In this lab, we learned how to use parallel circuits and how to use the current divider rule and Kirchoff's current law to calculate measurement and the current measurement. We had to calculate the total resistance to be approximately 162.4 ohms and the total current be 55.422 mA. The currents through the three resistors were 27.273 mA, 19.949 mA, and 9.000 mA, which added up to the total and current KCL. This lab helped learn on how to understand current divides through parallel resistors and how to properly measure using a DMM.